Cross-Entropy vs Mean Squared Error

Why Different Problems Need Different Losses

1. What a Loss Function Really Is

A loss function answers **one question only**:

> "How bad is the model's output for this input?"

Everything else in learning exists to **reduce this number**.

The loss:

- does **not** define the model
- does **not** define the data
- defines **what the model cares about**

2. Mean Squared Error (MSE)

Definition

$$L_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$$

What MSE Assumes

- Output is **continuous**
- Distance between values matters
- Large errors are much worse than small ones

What MSE Penalizes

- Predicting ₹10,000 instead of ₹40,000 → very large loss

- Predicting ₹39,000 instead of ₹40,000 → small loss

What MSE Ignores

- Probability
- Confidence
- Class boundaries

Where MSE Is Correct

- Salary prediction
- Price estimation
- Sensor readings
- Any numeric regression task

3. Why MSE Is Wrong for Classification

Consider a binary classification output:

- True label: `y = 1`
- Prediction: `ŷ = 0.51` vs `ŷ = 0.99`

Using MSE:

$$(0.51 - 1)^2 \approx 0.24$$

$$(0.99 - 1)^2 \approx 0.0001$$

Both are "correct class", but MSE treats them as **numerically different errors**, not **confidence differences**.

This is not what we want in classification.

4. Cross-Entropy Loss (Binary)

Definition

$$L_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\Big[y_i \log(\hat{p}_i) + (1-y_i)\log(1-\hat{p}_i)\Big]$$

Where:

- p^i∈(0,1)is predicted probability
- yi∈{0,1}

5. What Cross-Entropy Assumes

- Output represents **probability**
- Predictions close to 0 or 1 matter a lot
- Being confidently wrong is very bad

6. What Cross-Entropy Penalizes

If the true label is `1`:

- Predicting `0.99` → very small loss
- Predicting `0.6` → moderate loss
- Predicting `0.01` → **huge loss**

Cross-entropy strongly punishes **confident mistakes**.

This matches real classification risk.

7. Why Cross-Entropy Works with Sigmoid

Sigmoid outputs:

$$\hat{p} = \sigma(z) = \frac{1}{1+e^{-z}}$$

Properties:

- Output in `(0,1)`
- Interpretable as probability

- Smooth gradient everywhere

Cross-entropy + sigmoid produces **clean, stable gradients**.

This is not accidental.

They are mathematically matched.

8. Side-by-Side Comparison

Aspect	MSE	Cross-Entropy
Output type	Real number	Probability
Penalizes	Numeric distance	Confidence error
Sensitive to scale	Yes	No
Punishes confident mistakes	No	Yes
Used in	Regression	Classification
Activation	None	Sigmoid / Softmax

9. One Sentence Summary

MSE cares about how far you are.
Cross-entropy cares about how sure you were when you were wrong.

10. Connection Back to Labs

- **Lab 2 (Perceptron):** No loss, just correctness
- **Lab 3 (Logistic Regression):** Cross-entropy + sigmoid
- **Lab 4 (Linear Regression):** MSE + linear output

Same learning idea.

Different loss because the **question changed**.

> "Loss is not a technical detail.
>
> Loss defines what the model is trying to do."